



RESEARCH

Heat-tolerant algal symbionts may prevent extirpation of the threatened elkhorn coral, *Acropora palmata*, in Florida during intensifying marine heatwaves

Richard F. Karp¹ · Fabrizio Lepiz-Conejo¹ · Shayle B. Matsuda² · Bryce Corbett² · Alexandra D. Wen¹ · Joseph D. Unsworth¹ · Martine D'Alessandro¹ · Ken Nedimyer³ · Amelia Moura⁴ · Erinn M. Muller⁵ · Zachary Craig^{5,6} · Diego Lirman¹ · Ross Cunning^{2,1} · Andrew C. Baker¹

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Abstract The elkhorn coral, *Acropora palmata*, was historically a major reef-building species in the Caribbean, but has suffered devastating declines in recent decades. Despite significant restoration efforts in Florida, the marine heatwave of 2023 caused severe bleaching and mortality to both wild and restored colonies. To understand the disastrous impacts, we evaluated the variation in heat tolerance among Florida's *A. palmata* population prior to the event. In 2022, we used rapid acute heat stress assays to assess the thermal tolerance of 172 adult colonies (125 unique genets) from four nurseries. We found variation in thermal tolerance (4.17°C range in ED50) that was attributed to nursery location (17.2% of variation), genet (25.9%), and symbiont abundance (15.6%). Algal symbiont type, however, was the strongest predictor of

thermal performance, with the few ($n = 10$) colonies hosting *Durusdinium* being, on average, 1.9°C more thermally tolerant than corals hosting *Symbiodinium*. This difference would have decreased the effective heat stress accumulation during the 2023 event by ~92%. Therefore, despite considerable variation in thermal tolerance among Florida's elkhorn corals, hosting *Durusdinium* appears to be the most effective mechanism for surviving such extreme heat stress. These findings suggest that restoration strategies that focus on rearing sexually derived *A. palmata* recruits with *Durusdinium*, followed by outplanting to suitable environments, may improve survival during future heatwaves. Combined with efforts to introduce additional elkhorn diversity from populations outside Florida, these approaches may be the most effective interventions to promote the continued survival of Florida's elkhorn corals in the face of rapid climate change.

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✉ Richard F. Karp
rfk30@miami.edu

✉ Andrew C. Baker
abaker@earth.miami.edu

¹ Department of Marine Biology and Ecology, Rosenstiel School of Marine, Atmospheric, and Earth Science, University of Miami, 4600 Rickenbacker Causeway, Miami, FL 33149, USA

² Conservation Research Department, John G. Shedd Aquarium, 1200 South DuSable Lake Shore Drive, Chicago, IL 60605, USA

³ Reef Renewal USA, Tavernier, FL, USA

⁴ Coral Restoration Foundation, Key Largo, FL, USA

⁵ Mote Marine Laboratory, Sarasota, FL 34241, USA

⁶ Department of Land and Natural Resources, Division of Aquatic Resources, Kailua Kona, HI, USA

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Introduction

Ocean warming during July–September 2023 resulted in record high temperatures in the Caribbean and prolonged heat stress to coral reef ecosystems in the region (Hoegh-Guldberg et al. 2023). The accumulated heat stress severely impacted the Caribbean elkhorn coral, *Acropora palmata*, causing severe bleaching and mortality (NOAA 2024; Williams et al. 2024). Historically, *A. palmata* built foundational reef structure, providing essential fish habitat, coastline protection, and other economic benefits (NOAA Coral Reef Conservation Program 2013; Jackson et al. 2014; Ferrario et al. 2014; Beck et al. 2018; Toth et al. 2019). Since the 1950s, however, increased coastal development, *Diadema*

die-offs, disease outbreaks, and hurricanes reduced populations of *A. palmata* by at least 79% throughout the Caribbean (Lessios et al. 1984; Hughes 1994; Aronson and Precht 2001; Cramer et al. 2020). Lack of recovery through bleaching events, low genetic diversity, and asynchronous spawning led Williams et al., (2020) to project extirpation in Florida by 2030 in monitored plots (Miller et al. 2016; Williams et al. 2008, 2017, 2020). The 2023 marine heatwave, however, accelerated this prediction, with > 95% observed mortality of *A. palmata* across multiple outplant plots throughout the Florida Keys (NOAA 2024). Furthermore, Williams et al. (2024) found only 23 of 84 founder colonies (~27%) survived in the surveyed area of the upper Florida Keys, with only six of the 23 colonies (26%) appearing unaffected (Williams et al. 2024). Given the low survival rate, it is imperative to understand mechanisms of differential survival to help identify potential intervention strategies to increase the thermal tolerance of this valuable yet vulnerable species.

Algal symbiont type has been shown to be an important predictor of thermal tolerance (Baker et al. 2004; Fuller et al. 2020). Corals exist in symbiosis with a diverse range of dinoflagellate algae, which provide >95% of their metabolic energy and can alter key phenotypes such as thermal tolerance, growth rates, and reproductive output (Falkowski et al. 1984; Jones and Berkelmans 2010; Cunning et al. 2015; Fisch et al. 2019). In the Caribbean, associations with *Durudinium trenchii* can increase the thermal tolerance of the coral host by 1–2°C (Silverstein et al. 2015). *Acropora palmata*, however, is typically dominated by *Symbiodinium fitti* and only very rarely associates with *D. trenchii* (Baums et al. 2014; Kitchen et al. 2020). In addition to symbiont type, relative symbiont load (symbiont-to-host cell ratio) may also explain differences in bleaching susceptibility, with higher ratios leading to more severe bleaching (Cunning and Baker 2012; Cornwell et al. 2021). Therefore, key aspects of the symbiosis between *A. palmata* and its algal symbionts, such as associations with *Durudinium* or differences in symbiont load, may contribute to variation in thermal tolerance of this species across Florida's Coral Reef.

Environmental factors such as variation in temperature, light, depth, water quality, or water flow may further contribute to differential mortality across Florida's Coral Reef (Riegl and Piller 2003; Lesser and Farrell 2004; van Woesik et al. 2012; Smith et al. 2014; Fifer et al. 2021). Indeed, of the 23 surviving *A. palmata* colonies in the upper Florida Keys, 18 (78%) were found at a single reef site (Williams et al. 2024). Therefore, local environmental conditions such as cooler seawater or lower extreme temperatures may have contributed to higher survivorship (Williams et al. 2017, 2024). Reciprocal transplant studies of *A. cervicornis* during a natural bleaching event also demonstrated that outplant site significantly influenced bleaching severity for the same

genotype, which highlights that environmental conditions likely play a substantial role in modulating thermal tolerance (Drury et al. 2017; Fuller et al. 2020; Drury & Lirman 2021). Certain genets, however, were more thermally tolerant across all outplant sites, and thus certain genotypes may possess alleles which confer thermal tolerance that can be identified to understand what influences coral bleaching (Fuller et al. 2020; Drury and Lirman 2021).

Algal symbiont type, environment, and genetic factors can influence how a coral responds to heat stress (van Hooi-donk et al. 2016; Manzello et al. 2019; Fuller et al. 2020; Voolstra and Ziegler 2020). Assessing their interaction on thermal performance, however, is challenging due to difficulties standardizing across studies (Grottoli et al. 2021; McLachlan et al. 2021). Therefore, large comparative assessments of thermal tolerance have typically only been possible through field observations during natural bleaching events, which are subject to natural variability and confounding factors (Fuller et al. 2020). Coral Bleaching Automated Stress Systems (CBASS), however, have recently emerged as a standardized approach to experimentally quantify thermal tolerance and identify factors that underlie differences among individuals and populations (Voolstra et al. 2020; Cunning et al. 2021; Evensen et al. 2023). While acute stress tests may still need refinement to elucidate individual genetic performance, studies have shown these methodologies correlate to controlled short-term bleaching assay metrics of physiological response, population level thermal tolerance, and in some cases genet performance (Evensen et al. 2021; Klepac et al. 2024; Cunning et al. 2024). Thus, CBASS observations may be used to help explain bleaching field observations across a large range of genetic individuals. Therefore, we used data from CBASS assays undertaken in June 2022 to assess variation in *A. palmata* thermal tolerance and used this information to retrospectively understand the high levels of mortality on the reef following the 2023 marine heatwave through symbiont associations, nursery environment, and genetics.

Methods

Coral colony source locations

Thermal tolerance was assayed for *A. palmata* colonies being propagated on coral nursery trees (as described in Nedimyer et al. 2011) within in situ coral nurseries by the University of Miami (UM, colonies = 35), Coral Restoration Foundation (CRF, colonies = 78), and Reef Renewal (RR, colonies = 49). Colonies at these nurseries were originally collected from wild source colonies on the reef by restoration practitioners, with source colony latitude and longitude recorded at the time of collection with sources ranging from

Biscayne National Park to Key West (~200 km) and depths between 1.5 and 6.1 meters (Fig. 1). Colonies were maintained and propagated at in situ coral nurseries from the date of original collection (years: 2005–2021) until being collected for experimentation in June 2022.

Additional colonies from Mote Marine Laboratory (MML, colonies = 10) were assayed, but these were sexually derived individuals generated from batch crosses between 2013 and 2017 with parents in Biscayne National Park and Upper Keys (Brewster’s Reef, Elbow Reef, Horseshoe Reef, and Sand Island) that were grown and fragmented at Mote’s land-based facility (Summerland Key, Florida). Corals were maintained under controlled conditions in flow-through raceways (2.4 × 1.2 m) supplied with seawater sourced from an underground well at approximately 24 m depth, temperatures were controlled at ambient conditions based on season at approximately 29°C in the summer and 26°C in the winter, and irradiance was provided through natural sunlight with 50% shade cloth overhead. Water was processed through a sand filter and UV sterilization prior to utilization within the raceways. Corals were batch fed Golden Pearls twice a week.

Throughout this paper, colonies refer to a unique genet collected from the same coral nursery tree arm that are treated separately for statistical analysis. Since no genotype sequencing was conducted for this study, unique genetic individuals are referred to as genets and are distinguished

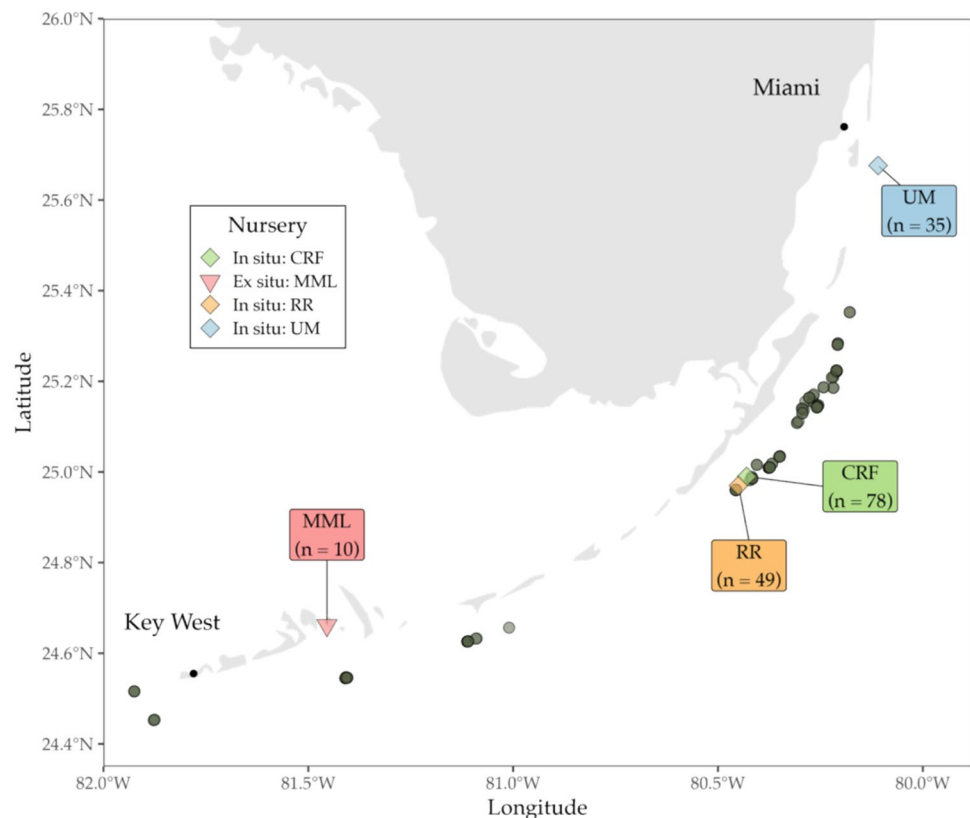
based on the provided nursery names. Since some genets were shared across nurseries and had different names based on the nursery, the Coral Sample Registry (as described in Moura et al. 2021) was used to match genets with the provided old name (“Old Genotype” metadata) to their new name for standardization across nurseries and allow for the comparison of genet in subsequent statistical analysis.

Coral collections and field operations

Heat tolerance assays were performed using CBASS set up on the deck of the R/V *Coral Reef II* in June 2022. At each in situ nursery, divers collected one to two fragments (larger coral pieces of a single colony to be subsequently cut into replicate ramets for CBASS) from available colonies (SI Table 1; n = 35–78 colonies per nursery). From Mote Marine Laboratory, eight replicate ramets (experimental coral piece that go in each temperature treatment) from ten genets derived from sexual recruits were driven by car from Summerland Key to Key Largo, Florida, in a water cooler and transferred to the R/V *Coral Reef II* after a two-hour drive. A total of 172 colonies were collected across four nurseries, comprising 125 unique genets, 39 of which were represented in multiple nurseries.

Fragments were held in the ship’s 8,000 L flow-through live well system prior to assays; corals were assayed within zero to 46 hours of collection. Approximately 1 hour before

Fig. 1 Map showing location of nurseries and the shape denoting nursery type (in situ or ex situ) with the location of original wild parent colony collection for *Symbiodinium*-containing corals in dark green



temperature ramping, fragments collected from in situ nurseries were divided into eight replicates approximately five cm² ramets using a diamond band-saw (Gryphon). Since Mote's corals were already provided as replicates on ceramic plugs, no further fragmentation was needed. One ramet per colony was distributed to each tank (30°C, 32°C, 33°C, 34°C, 34.5°C, 35°C, 36°C, and 37°C), so that each tank contained one ramet from all coral colonies in the run (n colonies = 35–48 per day). Corals were mounted on racks containing up to eight ramets, with maximum surface area facing upward. Racks were randomly distributed within each tank hourly to minimize the effect of tank position.

Heat stress assays

CBASS were constructed following the design of Voolstra et al., (2020) and Evensen et al., (2023) and set up as described in Cuning et al., (2021) with the following modifications: tank size was adjusted to 17-L, tanks were equipped with one titanium aquarium heater (Finnix TH-500 W), and four thermoelectric chillers (Nova Tec IceProbe) each. Water was supplied to each tank at a rate of 0.2 L/min (turnover = 1.4 hrs). Light was provided by LED aquarium lights (Phlizon 165 W) above each cooler and adjusted to provide 500 $\mu\text{mol photons m}^{-2} \text{ s}^{-1}$ at the center of each tank, measured with an underwater photosynthetically active radiation sensor (Apogee Instruments). All temperature profiles were conducted by custom controllers (ELEGOO Mega 2560) with temperature sensors (Vktech DS18B20) controlling heaters and chillers based on inputted set points. Temperatures in each tank throughout each run were recorded by the Vktech temperature probes and by independent HOBO pendants (Onset, MX2202).

Temperature profiles consisted of a 30°C baseline temperature, 3-hour ramp to a maximum temperature (ranging from 30°C to 37°C) at a rate of 0 to 2.3°C per hour based on treatment, a 3-hour hold at the target temperature, and 1-hour ramp down to 30°C at a rate of 0°C to 7°C per hour based on treatment temperature (Fig. S1a). Profiles were run so that the end of the high temperature hold coincided with local sunset and temperatures ramped down in the dark. After returning to 30°C over 1 hour in darkness starting at approximately 2000 hours, F_v/F_m was measured for each fragment at two points from non-overlapping areas of tissue using a MAXI-Imaging-Pam (Walz, Effeltrich, Germany). Specifically, the MAXI-Imaging-Pam was used to measure the background fluorescence (F_o) through the measuring light and maximal fluorescence (F_m) through the saturating pulse such that the variable fluorescence (F_v) could be calculated by subtracting F_o from F_m . To do this, fluorometer settings were as follows: measuring light intensity: 1, measuring light frequency: 1, damping: 1, saturation pulse intensity: 1, saturation pulse width: 0.120s. Gain was adjusted

between one to four so that background fluorescence (F_o) was above 0.130.

Symbiodiniaceae analyses

Tissue biopsies (approx. 0.5 cm²) were collected from each coral colony in the control (30°C temperature) treatment using curved nail clippers and preserved in a 1% SDS DNA buffer. DNA was extracted following a modified organic extraction protocol, and Symbiodiniaceae communities were quantified using genus-specific quantitative PCR (qPCR) assays targeting *Symbiodinium* and *Durisdinium* (Cunning and Baker 2012; Baker and Cuning 2016; Winter 2017). Symbiont-to-host cell ratios (S:H) were quantified by normalization to the single-copy CAM locus in *A. cervicornis* with copy number corrections for Symbiodiniaceae and no fluorescence corrections as PowerUp SYBR Green Master Mix (Applied Biosystems) was used to detect the presence of host and symbiont loci (Cunning & Baker 2012; Palacio-Castro 2019). For corals containing *Durisdinium*, Environmental Master Mix and TaqMan probes were used to identify the relative abundance of *Symbiodinium* and *Durisdinium* (Cunning and Baker 2012; Winter 2017). Therefore, fluorescence corrections were applied following Cuning and Baker (2012) and Winter (2017). Since qPCR assays were performed to the genus level, the symbiont types are referred by their genus name throughout this paper. Due to this study being conducted on *A. palmata* in the Caribbean, however, it is likely that *Symbiodinium* is *S. fitti* as *A. palmata* almost exclusively hosts this symbiont type in Florida and *Durisdinium* is *D. trenchii* as it is the only *Durisdinium* species in the Caribbean (Baums et al. 2014; Pettay et al. 2015).

Statistical analysis

Dose–response curves were used to model the decline in F_v/F_m as a function of the maximum treatment temperature for each genet. The treatment temperature was calculated as the average HOBO measurements during the 3-hour hold period in each tank, as there were small deviations from target values (Fig. S1b). Dose-response curves were fit as three-parameter log-logistic functions using the drc package (Ritz et al. 2015), with the following constraints on parameter values: maximum = [0.35, 0.65]; slope = [10, 120]; ED50 = [30, 50]. The lower limit was set equal to zero, and based on initial model fits, additional data filtering was performed to remove points with Cook's distance greater than 4/n as described in Cuning et al., (2021). Dose-response curves were then re-fit using filtered data to generate ED50 parameter values (and standard errors) for each colony. ED50 values are the temperature required to decrease photochemical efficiency by 50%, which are interpreted as the quantitative

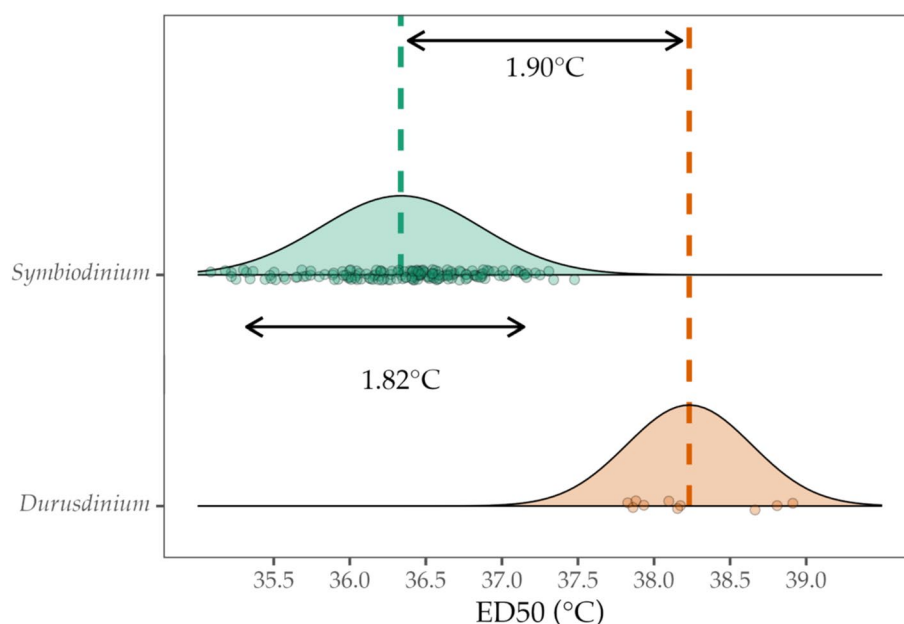
thermal tolerance phenotype for each colony and are the principal metric used in subsequent statistical analyses.

We used linear models to assess the variance explained by log-transformed S:H, nursery, dominant symbiont type, and genet on ED50. To partition the variance, we used a type I ANOVA to calculate the sequential sum of squares for each factor in the order that it was specified. Since dominant symbiont type was confounded with nursery (i.e., corals at MML hosted *Durusdinium*, while corals at all other nurseries hosted *Symbiodinium*), we fit two models to estimate the upper and lower bounds of variance explained by dominant symbiont type and nursery. In both models, log-transformed S:H ratio was specified first to account for its potential dependence on nursery and genet was specified last to quantify genetic variance only after environmental and symbiont effects were accounted for. Model 1 (log-transformed S:H → Nursery → Dominant symbiont type → Genet) estimated the upper bound of variance explained by nursery (since it was specified first) and the lower bound of variance explained by symbiont, while Model 2 (log-transformed S:H → Dominant symbiont type → Nursery → Genet) estimated the upper bound of variance explained by symbiont type and the lower bound of variance explained by nursery. To

Table 1 The variance explained by each factor on thermal tolerance (ED50) from Model 2 with an overall $R^2 = 0.949$

Variance explained (%)	
Symbiont load: log-transformed (S:H)	15.6
Dominant symbiont type	36.2
Nursery	17.2
Genet	25.9
Remaining variation	5.1

Fig. 2 Normal distributions of ED50s based on mean and standard deviation for each dominant symbiont type. Dotted lines show the means of the dominant symbiont (*Durusdinium* (orange) and *Symbiodinium* (green)). The arrow under *Symbiodinium* shows the range of the 5th–95th percentile thermal tolerance for *Symbiodinium* colonies, and the arrow between the two means shows the difference between the two dominant symbiont types



determine whether nursery (Model 1) or dominant symbiont type (Model 2) was the more likely driver of the high ED50 values observed in the MML corals hosting *Durusdinium*, we compared the mean ED50 difference between MML and all other nurseries (1.90°C) to the mean pairwise difference among nurseries hosting only *Symbiodinium* (0.59°C ± 0.35 s.d.). We then used a one-sample t-test to assess whether the high ED50 values from MML hosting *Durusdinium* reflect typical nursery variation or may be due to hosting *Durusdinium*. The statistically significant one-sample t-test (T-test, $t = 5.28$, $p < 0.05$) indicated that Model 2 and dominant symbiont type are likely a larger driver of thermal than nursery in this species. Due to this result, and because hosting *Durusdinium* is consistent with higher thermal tolerance in the literature, we report the variance explained by Model 2 in Table 1 and further discuss this variation throughout the manuscript (Berkelmans & van Oppen 2006; Silverstein et al. 2015).

In addition to using linear models to calculate variance partitioning, we performed subsequent pairwise tests to assess the significance of each factor. A Wilcoxon rank-sum test was then used to test for differences of ED50 based on dominant symbiont genera (Fig. 2).

Subsequent analysis was then conducted to determine the influence of symbiont load (symbiont-to-host cell ratio) and nursery on ED50. First, differences of log-transformed symbiont load and nursery were conducted using a Kruskal–Wallis test. Pairwise differences between ED50 and symbiont load due to nursery were then assessed using a pairwise Wilcoxon rank-sum test with a Bonferroni adjusted p-values ($\alpha = 0.01$) (Fig. 3a, c). Relationships between log-transformed symbiont load S:H and ED50 were performed independently for each dominant symbiont type and nursery

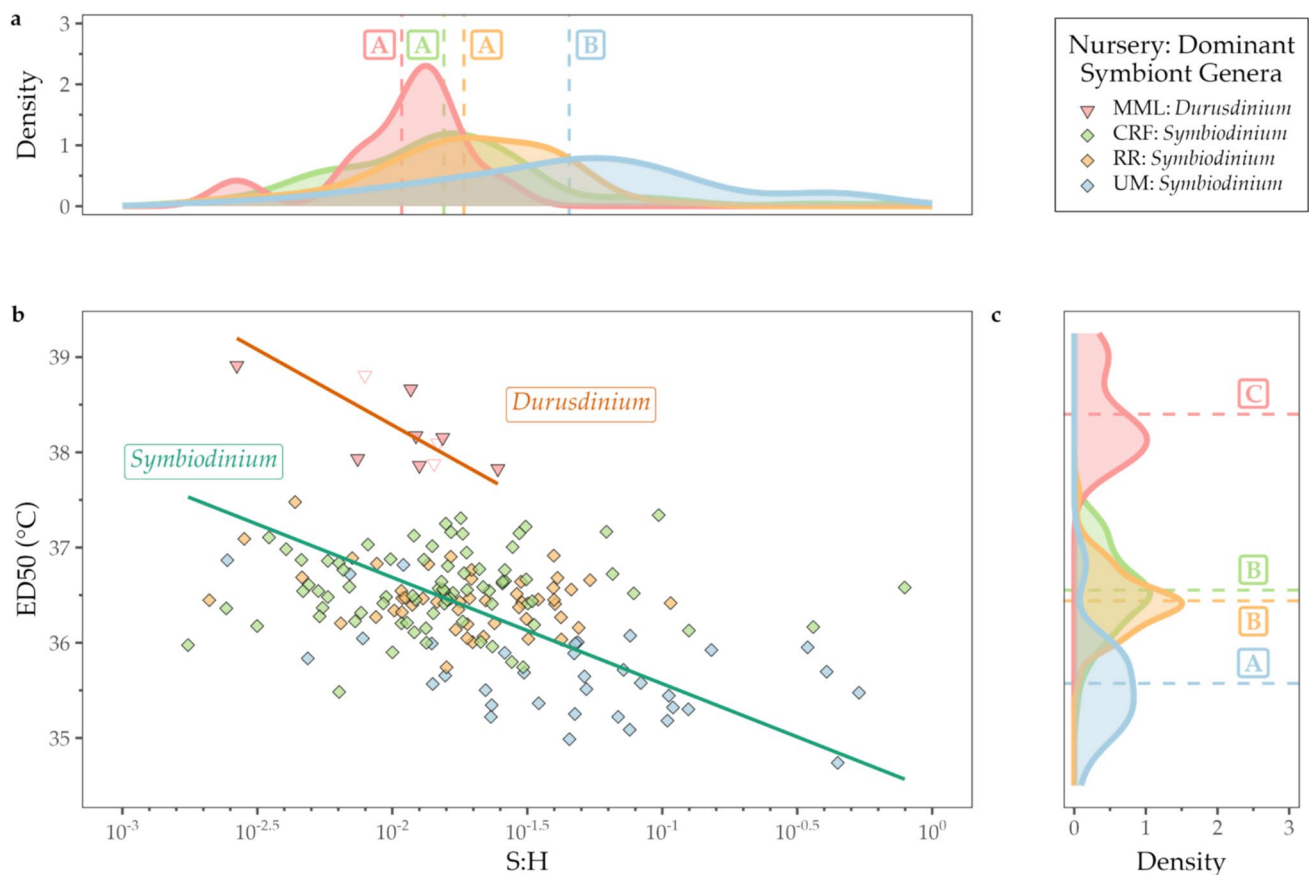


Fig. 3 **a** Density plots showing the distribution of S:H (symbiont-to-host cell ratio) colored by nursery. Letters denote significant pairwise differences at $\alpha = 0.01$ and dotted lines show nursery means. **b** Correlation of symbiont:host cell ratio (S:H) versus unadjusted ED50 with line of best fit as a Deming regression for each dominant symbiont type. The point shape denotes the dominant symbiont type and color denotes nursery. The three inverted triangles from MML with no inside color indicate the three colonies tested that

contained greater than 81%, but not 100% *Durussdinium*. The orange colored line depicts the Deming regression of *Durussdinium* ($\text{ED50} = -1.58 \cdot \log(\text{S:H}) + 35.12$) and the green colored line denotes the best fit line for *Symbiodinium* ($\text{ED50} = -1.12 \cdot \log(\text{S:H}) + 34.45$). **c** Unadjusted ED50s ($^{\circ}\text{C}$) for each nursery showing distribution, and statistical pairwise differences between nurseries shown as letters at $\alpha = 0.01$. The dotted line shows the overall mean for each nursery

using a Deming regression (permutations = 999), and the significance of the slope was tested using a t-test versus a null hypothesis slope of zero (Fig. 3b; Fig. S3). Lastly, variation in ED50 of all coral individuals at each nursery was visualized as a kernel density estimate and pairwise significance was conducted as described above (Fig. 3c).

To further assess genetic effects, which were measured at multiple nurseries, we calculated adjusted ED50 to standardize across the different environments. To do this, we used a third model to assess the main factors of log-transformed S:H and nursery environment on ED50 for only corals dominated by *Symbiodinium* (as they were the only genets shared across nurseries). We then added the residuals of this model for each coral to the mean for all corals measured dominated by *Symbiodinium*. The resulting ED50 adjusted distribution was tested for normality using a Shapiro-Wilk test and tested for the effect of the 39 genets, which were

measured across multiple sites using an ANOVA (Fig. S4). Pearson correlations were then performed on both ED50 and adjusted ED50 to test relationships of the same genet across nurseries (Fig. S5). Linear models were then used to test for relationships between ED50 and adjusted ED50 and source colony locations and the maximum monthly mean sea surface temperature for 5-km pixels containing original source colony coordinates, using Coral Reef Watch (CRW) CoralTemp V3.1 1985-2012 climatology (Fig. S6).

Finally, degree heating week calculations were performed on the 12-week rolling window at the 90th percentile hot spot on the Florida Keys 5-km Virtual Station (<https://coralreefwatch.noaa.gov/product/vs/timeseries/florida.php>) satellite sea surface temperature data with a maximum monthly mean (MMM) of 29.624 following Liu et al., (2003) (Liu et al. 2003, 2006). We then calculated the degree heating

week with an increased thermal threshold of $MMM + 0.90^{\circ}\text{C}$ and $MMM + 1.90^{\circ}\text{C}$ (Fig. 4).

All data analysis was conducted in RStudio, R version 4.2.1 (2022-06-23). All raw data and analysis is available at GitHub (<https://github.com/karprich/apalCBASS>) and archived at Zenodo (<https://doi.org/10.5281/zenodo.15121507>).

Results

Heat tolerance assays of 172 colonies of *A. palmata* from four nurseries with source parent colony collections spanning ~200 km across the Florida Keys (Fig. 1) revealed an ED50 range of 4.71°C ($34.74\text{--}38.91^{\circ}\text{C}$), with a mean of 36.45°C (± 0.68 s.d.) (Fig. 2). Coral fragment symbiont communities also varied in both symbiont type and abundance. The ten colonies from MML were dominated by *Durussdinium*: Seven hosted 100% *Durussdinium*, one was 99.9% dominant, another was 91.1% dominant, and the other contained 81% *Durussdinium*, with the remaining background symbionts being *Symbiodinium*. In contrast, the other 162 colonies from UM, CRF, and RR had 100% *Symbiodinium* (Fig. S2).

Dominant symbiont type explained the largest component of variation in ED50 (36.2%, Table 1). We found the corals dominated by *Durussdinium*, which were from MML, had a mean ED50 that was 1.90°C higher than corals with *Symbiodinium* from other nurseries (38.231°C versus 36.336°C) (Fig. 2, Wilcoxon, $p < 0.05$). To compare the magnitude of

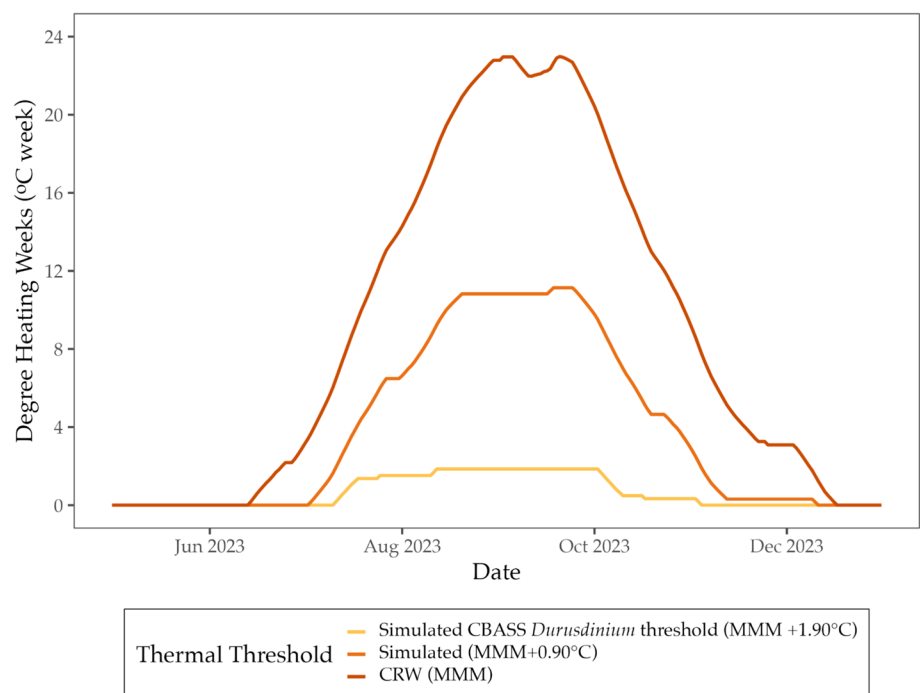
this difference to the range in heat tolerance among *Symbiodinium*-containing corals, the 90th percentile interval was calculated as 1.82°C .

While hosting *Durussdinium* and being from MML explained 36.2% of the variance, other factors directly associated with environment contributed to variation in ED50. We found symbiont load (measured as the total symbiont-to-host cell ratio, S:H) varied among colonies with a 2.65-fold range, and mean S:H varied across nurseries (Fig. 3a). These differences in symbiont load were negatively correlated with ED50, regardless of their dominant symbiont type, and explained 15.6% of overall variance in thermal tolerance (Table 1, Fig. 3b, T-test, $p < 0.05$). The higher the symbiont:host cell ratio, the lower the ED50, such that a doubling of S:H led to a decrease in ED50 of 0.34°C and 0.48°C for *Symbiodinium* and *Durussdinium*-dominated corals, respectively. This relationship was observed across nurseries, but also within all nurseries except CRF (Fig. S3).

Although differences in the average symbiont load among nurseries partly explained differences in their average ED50s, other nursery (e.g., environmental) factors explained an additional 17.2% (Table 1) of the variance in ED50. Comparing between nurseries with *Symbiodinium* dominance, CRF and RR had similar average ED50 values of 36.56°C and 36.46°C , respectively, but UM had a significantly lower mean value (35.67°C ; Fig. 3c, $p < 0.05$, Kruskal–Wallis).

Most of the remaining variation in ED50 was attributed to genet (25.9%, Table 1) with the remaining model variance (5.1%, Table 1) being attributed to measurement error. To compare the contribution of only genetic effects across

Fig. 4 Accumulated degree heating weeks at the 90th percentile hot spot across the Florida Keys from the Florida Keys 5-km Virtual Station, showing the actual degree heating week accumulation under the standard bleaching threshold for the Florida Keys maximum monthly mean (MMM) of 29.6264 (red), as well as heat stress accumulation under a thermal threshold increase of $MMM + 0.90^{\circ}\text{C}$ (orange), and the thermal threshold increase of $MMM + 1.90^{\circ}\text{C}$ estimated using CBASS for *A. palmata* hosting *Durussdinium* (yellow)



multiple environments, we also calculated adjusted ED50 to standardize for both nursery and symbiont load. We then compared the ED50 and adjusted ED50s for the 39 genets that were represented at multiple nurseries (32 genets measured at 2 nurseries, and 7 genets measured at 3 nurseries). Genet was a significant predictor of both ED50 and adjusted ED50 (Fig. S4, $p < 0.05$, ANOVA) with the most thermally tolerant individuals measured at multiple sites being on average 1.58°C or 1.41°C more thermally tolerant through ED50 or adjusted ED50, respectively. Additionally, the same genets represented across multiple nurseries showed a significant correlation between two of the three nursery combinations for both ED50 and adjusted ED50 (Fig S5).

ED50 and adjusted ED50 were then compared to latitude, longitude, and maximum monthly mean temperature from satellite-derived sea surface temperatures to correlate thermal tolerance to original colony sources. We found no correlation between latitude, longitude, or mean monthly maximum temperature (Fig. S6, $p > 0.05$, Pearson).

Discussion

This study represents the first census of thermal tolerance for the elkhorn coral *Acropora palmata* in Florida. The range in thermal tolerance among colonies hosting *Symbiodinium* was 2.73°C (34.74–37.47°C), similar to the range of 2.57°C found in Floridian *A. cervicornis*, which also host *Symbiodinium* (35.05–37.62°C) (Cunning et al. 2021). However, when *A. palmata* hosted predominantly *Durudinium* (> 81%) and were from MML, their thermal tolerance increased by 1.90°C on average. The large increase in thermal tolerance due to hosting *Durudinium* is greater than the 90th percentile interval of corals hosting *Symbiodinium* (1.82°C). Additionally, the increase from hosting *Durudinium* was ~3.2 times greater than the average difference between nurseries hosting *Symbiodinium*. These results strongly suggest that hosting *Durudinium* can increase thermal tolerance more effectively than either coral genetics or environmental conditions, at least given the remaining genetic diversity of *A. palmata* available in Florida that we measured hosting *Symbiodinium*.

Indeed, previous estimates of the additional heat tolerance imparted by *Durudinium* through conventional longer-term bleaching assays have shown that *Durudinium* increases thermal tolerance by 1–2°C, which is similar to the 1.90°C reported here using CBASS (Berkelmans & van Oppen 2006; Silverstein et al. 2015). Consequently, despite significant differences in approach, the rapid heat profiles used by CBASS appear to reach similar conclusions for increases in thermal tolerance, as previous methods.

Although hosting *Durudinium* and being reared at MML was the main driver of thermal tolerance increase in *A. palmata* (explaining 36.2% of variance) in the present study, additional factors also influenced thermal response. Environment (nursery site) contributed 17.2% to the variation in thermal tolerance, but increased to 31.1% for colonies only hosting *Symbiodinium* and those measured directly from in situ coral nurseries. Therefore, recent environmental history can significantly alter measured thermal tolerance, which may be driven by differences in temperature regimes, turbidity, irradiance, or water quality (Palacio-Castro et al. 2022; de Souza et al. 2023). However, because coral colonies in this study were continuously propagated at in situ or ex situ nurseries for on average greater than five years, we are unable to identify the exact environmental drivers, which led to this large variation in thermal tolerance. Nevertheless, environmental conditions may play a crucial role in coral survival as surveys after the 2023 marine heatwave found that 18 (78%) of the 23 surviving *A. palmata* colonies in the upper Florida Keys were found at a single reef site (Williams et al. 2024).

Differences in environmental conditions may further alter symbiont load, symbiont and microbial communities, and/or host tissue properties, which can all contribute to differences in thermal tolerance (Palacio-Castro 2019; Palacio-Castro et al. 2022; van Woesik et al. 2022). Indeed, we found that nursery environment significantly affected symbiont load (Fig. 3a), which was itself a significant predictor of ED50 that explained 15.6% of the variance (Fig. 3b). This finding aligns with previous longer-term bleaching studies showing that corals with higher symbiont loads are more susceptible to bleaching (Cunning and Baker 2012; Cornwell et al. 2021). Therefore, although multiple environmental factors can directly affect thermal tolerance, its greater contribution may be the environment's ability to alter both symbiont type and load, which account for a majority of thermal tolerance in this species.

Genetic effects explained an additional 25.9% of the overall variance in thermal tolerance, which is a greater contribution than symbiont load or environment (nursery site). This enabled us to identify several thermally tolerant or sensitive individuals across multiple environments hosting *Symbiodinium* (Fig. S4). These results are similar to field bleaching studies in *A. cervicornis*, which have identified thermally tolerant individuals across multiple environments (Drury et al. 2017; Million et al. 2022).

The relative contributions of genet, environment, and symbiont type to the overall variation in thermal tolerance may help explain the difficulty in identifying molecular markers of heat tolerance (Fuller et al. 2020; Drury and Lirman 2021). This is a similar finding to Fuller et al., (2020), who highlighted through genetic analysis on *Acropora millepora* the influence of these compounding factors. Therefore,

to identify molecular markers or thermally tolerant genets, many genets need to be sourced from multiple different environments to control for the interacting contributions of environment, symbiont type, and load on the coral host. Similarly, for corals with the same symbiont type, nursery environment may have a comparatively large effect on thermal response (likely due to its effect on symbiont load), thereby further obscuring the effect of host genetics.

Furthermore, we found no correlation between thermal tolerance and the original reef site from where these nursery colonies were collected (latitude, longitude, or maximum monthly mean temperature). This is similar to the findings of Cuning et al., (2021) and Cornwell et al., (2021) that identified thermally tolerant individuals of *A. cervicornis* and *A. hyacinthus*, respectively, at multiple sites across an entire population. However, due to the relatively small scale over which we assessed thermal tolerance (~200 km), we cannot rule out the likelihood that corals sourced from very distant populations (outside Florida) may vary significantly in their thermal tolerance based on their origin, with latitude and temperature regime likely to be significant drivers of that variation (Dixon et al. 2015; Evensen et al. 2022).

The marine heatwave of 2023 also devastated wild and outplanted populations of *A. cervicornis*. We compared our results for *A. palmata* from four nurseries to those of Cuning et al., (2021), for *A. cervicornis* from seven nurseries, including the same four nurseries (CRF, MML, RR, and UM see SI Table 2). When comparing within only colonies containing *Symbiodinium* as no colonies contained *Durisdinium* in Cuning et al., (2021), the effect of the nursery environment was remarkably similar between *A. palmata* and *A. cervicornis*, with a similar rank order, variance explained by nursery environment (31% vs. 37%), and standard deviation of adjusted ED50s (0.38 versus 0.39; SI Table 2). Taken together, *A. cervicornis* and *A. palmata* may have a similar thermal tolerance range and standard deviation. Therefore, identifying mechanisms that influence thermal tolerance of *A. palmata* may also have immediate applicability for *A. cervicornis*. Since *A. palmata* and *A. cervicornis* have seen similar declines throughout the Caribbean, share similar species and life history traits, and are heavily used in restoration, identifying intervention strategies in either species may accelerate progress in the other species (McWilliam et al. 2018; Cramer et al. 2020; Boström-Einarsson et al. 2020).

Multiple intervention strategies have been proposed to increase corals' resilience to warming oceans (van Oppen et al. 2015; National Academies, 2019; Bay et al. 2019; Voolstra and Ziegler 2020). Selective breeding and assisted gene flow (AGF) are intervention strategies that can both increase genetic diversity and thermal tolerance by selecting heat-tolerant individuals (National Academies, 2019; Baums et al. 2019; Bay et al. 2023). Algal symbiont manipulations have been proposed as a separate avenue to increase

thermal tolerance and can be scaled by provisioning corals with appropriate algal symbionts in early life stages (van Oppen et al. 2015; Williamson et al. 2021; Coffroth et al. 2022; Nitschke et al. 2024). Here, we identified the relative contribution of the underlying factors that influence thermal tolerance for these interventions to better prioritize strategies to increase thermal tolerance in *A. palmata*.

Thermal tolerance measurements using CBASS show a wide range of both ED50 (2.73°C) and adjusted ED50 (2.18°C) across colonies hosting *Symbiodinium*, which suggests large differences in thermal tolerance among *A. palmata* genets that can be driven by either environmental or genetic factors. This variation, however, may have limited ecological value during marine heatwave events in Florida, as evidenced by the widespread mortality of *A. palmata* across Florida's Coral Reef, particularly in the central and southern Florida Keys (Williams et al. 2024). Therefore, it may be necessary to introduce additional host genetic diversity from outside Florida to bolster local populations, which may be more thermally tolerant, especially if sourced from warmer habitats.

Alternatively, if *A. palmata* can be manipulated to host and retain *Durisdinium*, these corals may maintain a competitive advantage during stressful ocean temperatures as it may directly increase their thermal tolerance. Indeed, if this studies measured increase for hosting *Durisdinium* (1.90°C) is applied to the thermal threshold for the Florida Keys at the 90th percentile hot spot sea surface temperature and degree heating weeks for the 2023 marine heatwave are recalculated, the effective heat stress accumulation is reduced from 23.0 to 1.85°C-weeks—a reduction of 92% (Fig. 4). Additionally, even an increased bleaching threshold of 0.90°C would reduce effective heat stress to just 11.1°C-weeks (a 52% decrease in heat stress accumulation). Therefore, hosting *Durisdinium* would likely have significantly improved bleaching and survival on the reef, even through the unprecedented marine heatwave of 2023.

This study suggests that, in addition to sourcing additional *A. palmata* diversity from outside Florida, an alternative strategy for increasing heat tolerance in restored *A. palmata* is to provision them with *Durisdinium*. This coral-algal association is very rare in the Florida Keys, and in this study, the only colonies of *A. palmata* hosting *Durisdinium* were the ten colonies from MML that were raised from sexual recruits in a land-based facility in Florida. This makes it difficult to disentangle the exact effect of symbiont type versus grow out conditions and genetic influences as neither the same genets nor symbiont type were present at the other in situ nurseries assessed. Therefore, the observed increase in thermal tolerance for the corals hosting *Durisdinium* from MML could be due to multiple interacting factors and not dominant symbiont type. For instance, the MML corals were five to nine-year-old corals derived from sexual recruits,

which had been repetitively fragmented into small ramets ($\sim 5 \text{ cm}^2$). This may make these fragments more resilient to thermal stress than the colonies measured from the other in situ nurseries that had been directly collected from wild colonies as Miller et al. (2024) showed that assisted sexual recruits showed higher bleaching resistance throughout the 2023 marine heatwave. Similarly, rearing in a land-based nursery may have caused these corals to have different physiological metrics compared to wild populations and therefore perform differently than colonies taken directly from the in situ nurseries containing *Symbiodinium* (Gantt et al. 2023). Indeed, studies by Gantt et al., (2023) showed that land-based *A. palmata* had increased chlorophyll *a* concentrations, but lower tissue biomass compared to wild coral colonies, which could contribute to altered thermal tolerance of MML corals versus corals from in-water nurseries. Additionally, just as nursery environment can alter the symbiont community, it may also influence microbial taxa, which could influence thermal tolerance (Maher et al. 2020; Baker et al. 2022; Palacio-Castro et al. 2022). Finally, although the genets from MML were not represented at any of the other nurseries, they were offspring from adult colonies in the upper Florida Keys and Biscayne National Park, which are from the same population as the genets measured hosting *Symbiodinium* and suggests that the genets from MML were likely not genetically distinct (Baums et al. 2005). While differences in environment, life stage, and genetics, may have also contributed to increases in thermal tolerance for the MML corals, the fact that the colonies from MML were all dominated by *Durusdinium* and showed significant increases in thermal tolerance that are similar to those reported for other coral species, strongly supports that *Durusdinium* dominance was somewhat responsible for the increased heat tolerance of these individuals.

Since all of the MML corals were dominated by *Durusdinium* and other studies on MML corals have found similar symbiont associations (e.g., Merck et al. 2022; Gantt et al. 2023), it is likely that symbiosis with *Durusdinium* in acroporids can be altered due to their initial rearing conditions. Thus, the symbiosis between *A. palmata* and *Durusdinium* can be stable in these land-based facilities and subsequent field monitoring studies by Elder et al., (2023) have shown that *Durusdinium* associations can be maintained for at least two years after outplanting to the reef (Merck et al. 2022; Elder et al. 2023). This association, however, may not last long term at all the reef sites where these corals are outplanted, and consequently, corals may lose their thermal tolerance due to symbiont community shifts (Gantt et al. 2025). Indeed, out of thousands of MML outplants along Florida's Coral Reef, only a few outplants survived the summer 2023 marine heatwave, which could have still been hosting *Durusdinium* or may have experienced less localized thermal stress due to their environment (E. Muller,

personal communication). Future research will need to focus on understanding the environmental drivers that maintain *Durusdinium* in Florida's acroporids, which may suggest alternative sites (e.g., warmer or inshore) for outplanting where these associations may show greater longevity.

This study found that, although considerable variation in heat tolerance exists in Florida's *A. palmata* population, this variation was demonstrably insufficient for this species to survive the unprecedented marine heatwave of 2023, unless associations with heat-tolerant symbionts are considered. Therefore, selective breeding from only Floridian corals hosting *Symbiodinium*, which were heavily impacted during the 2023 marine heatwave, may be unlikely to substantially increase heat tolerance when faced with similar bleaching events unless additional diversity from outside Florida with higher thermal tolerance is introduced. Therefore, understanding how and why Caribbean acroporids can establish and maintain symbioses with *Durusdinium* may be critical to developing strategies to enhance their survival. Thus, prioritizing the production of sexual recruits that host *Durusdinium*, paired with studies to understand the longevity and the trade-offs of these associations in different outplanting environments, represents an actionable intervention to significantly increase the thermal tolerance of restored *A. palmata*. This strategy, in addition to the introduction of additional host genetic diversity from suitable nearby populations, especially those in warm environments, is critical to maximize the ability of Florida's *A. palmata* to survive continued marine heatwaves that are likely to become common in a new era of extreme warming.

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Author Contribution R.K. contributed to conceptualization, methodology, validation, formal analysis, investigation, data curation, writing-original draft, visualization. F.L., S. M., B.C., and A.W. contributed to investigation. J.U., M.D., K.N., A.M., E.M., Z.C., and D.L. contributed to resources. R.C. and A.B. contributed to conceptualization, methodology, validation, visualization, supervision, and funding acquisition. All authors reviewed and edited the manuscript.

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Data availability All raw data and analysis code are available at GitHub (<https://github.com/karprich/apalCBASS>) and archived at Zenodo (<https://doi.org/10.5281/zenodo.15121507>).

Declarations

Conflict of interest The authors declare no competing interests.

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